Independent disaster-plan analysis

This is an independent analysis, not government approval or an operational order.

Report: evaluated

Source: Guiguinto MDRRM Plan 2023-2025
SHA-256: b355207cefdee01f3a67876b1f11b442b90c1acb7b7c6b49ad48433c29aafb95

Modeled scope: Poblacion-centered Guiguinto pilot cell; exact model-cell polygon ((120.874, 14.823), (120.884, 14.823), (120.884, 14.832), (120.874, 14.832), (120.874, 14.823)); one hypothetical 100-person cohort; no full jurisdiction estimate

Exclusions

- Rest of Guiguinto: Outside this bounded pilot cell and cohort
- Only the named 100-person hypothetical cohort is modeled; no plan-wide population is allocated
- Neighboring facilities and populations are excluded; their availability and permission are unknown
- The rectangle is a modeling cell, not an administrative barangay boundary
- Road-node proximity is checked; physical connector access and shelter operability are assumptions

Source and supplemental evidence

Field	Source value	Unit	Citation
municipal_population_context	112889	PERSON	Original PDF physical page 38
flood_exposed_people	56516	PERSON	Original PDF physical page 80
flood_exposed_households	11300	FAMILY	Original PDF physical page 80
shelter_capacity_people	3987	FAMILY	Original PDF physical page 121
shelter_capacity_people	4087	FAMILY	Original PDF physical page 121
pickup_policy	barangay halls	TEXT	Original PDF physical page 122
primary_center_context	GMACC	TEXT	Original PDF physical page 164
vehicle_seats	600000	PESO	Original PDF physical page 136
guiguinto_population	The 2020 census population is 112,889 across 14 barangays. It is not the same as flood exposure or scenario evacuation demand.	source text	Original PDF physical page 38
guiguinto_water_context	In June 2014, 11,861 of 20,194 households were reported with Level 3 water connections; the plan separately mentions 23 pumping stations. Neither is a measured initial shelter water stock.	source text	Original PDF physical page 46
guiguinto_flood_exposure	The flood section lists 56,516 affected people and 11,300 affected households. These overlap the municipality population and are not added to it.	source text	Original PDF physical page 80
guiguinto_visual_area	The visually inspected table prints 11.43 square kilometres under high flood susceptibility and 10.01 under low-to-moderate susceptibility; these are areas, not people or evacuation demand.	source text	Original PDF physical page 84
guiguinto_shelter_units_range	The split table labels school, day-care and covered-court capacities in FAMILY units. Day-care entries use the 5-6 range; no family-to-person factor is supplied by this table.	FAMILY	Original PDF physical page 120, 121
guiguinto_capacity_discrepancy	Printed capacity subtotals 2,753 + 84 + 400 + 500 + 50 + 200 sum to 3,987 FAMILY, while the printed grand total is 4,087 FAMILY. Both values are retained as a critical conflict.	source text	Original PDF physical page 121
guiguinto_shared_facilities_and_marks	Page 121 uses graphic checks/crosses for facility presence and prints GMACC, Kanlungan and Municipal Hall capacities in shared columns. Provider assignments of those shared capacities to Cutcut are rejected pending location/capacity review; blank cells are unknown, not zero.	source text	Original PDF physical page 120, 121
guiguinto_pickups_modes	Barangay halls are stated pickup points, with barangay/municipal service vehicles and walking or small/medium vehicle access described. Seats, exact schedules and road times are not established.	source text	Original PDF physical page 122
guiguinto_proposed_resources	The table is headed Proposed/Programs/Projects/Activities and includes rescue vehicle/boat procurement costs. The provider mislabeled currency values as COUNT; those candidates are rejected and cannot become fleet inventory.	source text	Original PDF physical page 136
guiguinto_gmacc_role	The plan calls GMACC the primary evacuation center. Its exact coordinate, access connector and reconciled capacity are not supplied by that sentence.	source text	Original PDF physical page 164
reviewed road graph	ef85554a12909a7d0728ee1ca390718714b2c721cf73cec04e222435fd0f9500	SHA-256	release/validation/geography/guiguinto-small https://download.openstreetmap.fr/extracts/a

Readiness gaps

- evacuee_people: The document does not establish this input. Review a scoped cohort count or declare a bounded modeling assumption. Depends: evacuated population, arrival time
- shelter_capacity_people: Competing source values need a recorded review decision. Resolve facility identity and FAMILY units, or declare a reviewed capacity assumption. Depends: shelter feasibility, unserved population
- vehicle_seats: Only proposed or target resources are documented. Supply current fleet seats or declare a scenario assumption. Depends: trip feasibility, arrival time
- opening_water_ml: The document does not establish this input. Supply dated stock evidence or declare a scoped water assumption. Depends: water sufficiency, water shortfall
- Procedure not modeled: walking
- Procedure not modeled: staged transfers
- Procedure not modeled: resupply

Explicit assumptions

- Personally reviewed local model cell and OSM connectors; physical access and facility operation assumed
- One hypothetical 100-person cohort, 20-seat prepositioned vehicle, one dedicated crew, 150-person shelter with zero initial occupants; 80 people remain at pickup
- Opening water: pickup 100,000 mL and shelter 400,000 mL; 2,000 mL/person/day over 259,200 seconds. Hypothetical change adds 500,000 mL at pickup
- Passenger deadline 70 seconds; one second positioning, loading and unloading each
- Road closure at time zero on exact edge osm:1341516497:0:forward; known to policy at time zero
- Car-road routing uses a declared 30 km/h fallback where OSM speed is unknown; the selected anchors and route do not establish field travel or accessibility
- No FAMILY-to-PERSON conversion; city/municipal historic totals and proposed resources are context only
- Variant is hypothetical and untested in the field; no resupply during the three-day horizon

Measured outcomes

Metric	Value	Unit
normal baseline on-time arrival	20	people
normal baseline unmet water	380031	mL
normal car route travel	64	seconds
road_close baseline on-time arrival	0	people
road_close baseline unmet water	380034	mL
road_close car route travel	71	seconds
normal added-water on-time arrival	20	people
normal added-water unmet water	0	mL
normal car route travel	64	seconds
road_close added-water on-time arrival	0	people
road_close added-water unmet water	0	mL
road_close car route travel	71	seconds

Regressions

- No regression under the recorded four-metric paired comparison

Attribution

- Independent exploratory analysis; hypothetical assumptions are not plan facts
- Road geometry © OpenStreetMap contributors, ODbL 1.0; regional snapshot 2026-10-13
- No field predictive validation or government approval is claimed